

Evidence boundary after formal robustness analyses

GSE127136 within-cell-type UCell	Localized	No FDR-stable difference	Not established
GSE127136 pseudobulk modules	Effect directions	No FDR-stable difference	Not established
GSE286911 formal UCell direction	Complete pool rank separation	Sampling-stable; zero-inflated	Not established
GSE220100 restricted RIPS	Group contrast	Normalization direction reversal	Not established
GSE220100 complement module	FDR signal in one path	Normalization-specific only	Not established
Observed signal Robustness result Disease-specific inference			

Clinical biomarker, prognosis, treatment selection, and causal claims remain unsupported.